Teacher & Parent Guide

This workbook teaches computational thinking through 12 unplugged activities aligned to CSTA K-12 CS Standards. Each activity builds on the previous ones, and every activity has both a Basic (ages 8-10) and Challenge (ages 10-12) version for natural differentiation. Use this guide to plan lessons, track standards coverage, or extend the material.

Learning Objectives by Activity

#	Activity	Learning Objective	CSTA Standard
1	**What is an Algorithm?**	Understand that algorithms are step-by-step instructions	1B-AP-08
2	**Sequencing Puzzle**	Recognize that order matters in algorithms	1B-AP-10
3	**Spot the Loop**	Identify repetition in real-world sequences	1B-AP-09
4	**Write Your Own Loop**	Use loops and nested loops in pseudocode	1B-AP-11
5	**Yes/No Conditionals**	Apply IF/THEN/OTHERWISE logic to decisions	1B-AP-11
6	**Nested Conditionals**	Chain conditionals into decision trees	1B-AP-11
7	**Debug the Robot**	Identify and fix bugs in instruction sets	1B-AP-15
8	**Bubble Sort**	Execute a named sorting algorithm unplugged	1B-AP-08, 09
9	**Selection Sort**	Compare two algorithms for the same task	1B-AP-08, 12
10	**Pattern Recognition**	Identify repeating, growing, and rule-based patterns	1B-AP-10
11	**Treasure Hunt Grid**	Integrate sequencing, loops, and conditionals	1B-AP-11, 15
12	**Design Your Own Algorithm**	Create original algorithms from scratch	1B-AP-11, 14

CSTA K-12 Standards Reference

1B-AP-08 *(Modularity)* — Compare multiple algorithms for the same task.
1B-AP-09 *(Algorithms)* — Create programs that use algorithms to solve problems.
1B-AP-10 *(Control)* — Create programs that include sequences, events, loops, and conditionals.
1B-AP-11 *(Modularity)* — Decompose problems into smaller, manageable subproblems.
1B-AP-12 *(Program Development)* — Modify, remix, or incorporate portions of an existing program into one's own work.
1B-AP-14 *(Modularity)* — Give attribution when using ideas and creations of others.
1B-AP-15 *(Program Development)* — Test and debug a program or algorithm to ensure it runs as intended.

How to Use This Workbook

- Introduce each activity by reading the "How It Works" section aloud. This front-loads the key concept before kids attempt the exercises.
- Let kids attempt the Basic version independently, then transition to Challenge only if they're ready. Not every kid needs to do the Challenge — that's the point of differentiation.
- Use the Bonus activities for extension work, homework, or "early finisher" tasks. Many are hands-on or social.
- Return to earlier activities anytime — the material is designed to compound, so review is genuinely valuable, not remedial.
- Discussion prompts in each Teacher/Parent Note give you conversation-starters ready to go.

Extension Ideas

1. Coding Playground

Introduce kids to Scratch (scratch.mit.edu) or Code.org's free courses. The concepts they learned here — loops, conditionals, sequences, decomposition — map directly to visual coding blocks. Most kids will feel instantly at home.

2. Algorithm Journaling

Have kids identify one algorithm they encounter each day for a week — a school-day routine, a game strategy, a household process. Discuss what tools it uses (sequence, loop, conditional). This turns pattern recognition into a lifelong habit.

3. Teach-Back Project

Have your child teach ONE activity from this workbook to a younger sibling, friend, or classmate. Teaching is the deepest form of learning — and it's the same activity that closes this workbook (Activity 12 Bonus).

4. Design Their Own

Encourage kids to design a workbook page of their own — inventing a new activity that teaches an algorithm concept. Kids who can design lessons truly understand the material.

Questions or feedback? We'd love to hear from you at yusufahmed.sdet@gmail.com or on LinkedIn (linkedin.com/in/md-yusuf-ahmed).